Measurements and Instrumentation (BECS 22443)

Chapter 1
Digital Logic Families

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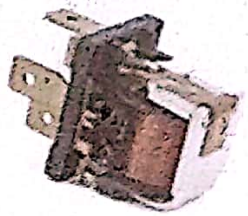
REVIEW

Introduction to Logic Gates

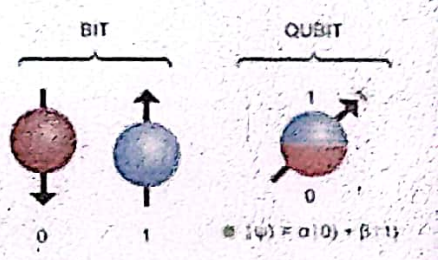
- In the earliest computers, switches were actuated through the manipulation of magnetic fields, achieved by energizing coils within **relays**. These switches, in turn, controlled the flow of electrical current.



- Subsequently, electronic **vacuum tubes** took over the role of relays in opening and closing current paths.

- Today, **transistors** serve as electronic switches, opening and closing current paths.

- In the future, **quantum bits, or qubits**, will operate as novel switches, leveraging the principles of superposition and entanglement for unprecedented computational capabilities



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Basics Gates

The basic logic gates NOT, AND and OR are fundamental to digital circuits. They perform simple binary operations and serve as the essential building blocks for more complex logic systems.

❑ The NOT Gate

Symbol	Truth Table	
	A	Q
	0	1
Inverter or NOT Gate	1	0

Boolean Expression Q = not A or $\overline{A}$ = Inverse of A (Line on Q)

❑ The AND Gate

Symbol	Truth Table		
	B	A	Q
	0	0	0
	0	1	0
	1	0	0
2-input AND Gate	1	1	1

Boolean Expression Q = A.B | A AND B gives Q

❑ The OR Gate

Symbol	Truth Table		
	B	A	Q
	0	0	0
	0	1	1
	1	0	1
2-input OR Gate	1	1	1

Boolean Expression Q = A+B | A OR B gives Q

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Universal gates

Universal gates, such as NAND and NOR, play a crucial role in digital electronics due to their capability to replicate the functions of all other fundamental logic gates. This inherent versatility enables more efficient circuit design by minimizing the necessity for various gate types. Consequently, universal gates are extensively utilized in the construction of complex digital systems.

❑ The NAND Gate

Symbol

2-input NAND Gate

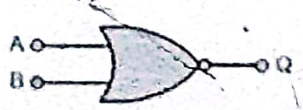
Truth Table

B	A	Q
0	0	1
0	1	1
1	0	1
1	1	0

Boolean Expression $Q = \overline{A.B}$ | A AND B gives NOT Q

❑ The NOR Gate

Symbol



2-input NOR Gate

Truth Table

B	A	Q
0	0	1
0	1	0
1	0	0
1	1	0

Boolean Expression $Q = \overline{A+B}$ | A OR B gives NOT Q

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Special Purpose Gates

Special purpose gates are designed to perform specific functions that go beyond the capabilities of standard logic gates (AND, OR, NOT). These gates are integral to enhancing the functionality of digital circuits, enabling more complex operations and facilitating efficient design in various applications.

❑ The X-OR Gate

Symbol

2-input Ex-OR Gate

Truth Table

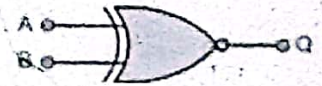
B	A	Q
0	0	0
0	1	1
1	0	1
1	1	0

Boolean Expression $Q = A \oplus B$ — A OR B but NOT BOTH gives Q

- It may be noted that if both inputs of the X-OR gate are high or both inputs are low, then the output is low

❑ The X-NOR Gate

Symbol



2-input Ex-NOR Gate

Truth Table

B	A	Q
0	0	1
0	1	0
1	0	0
1	1	1

Boolean Expression $Q = \overline{A \oplus B}$ — If A AND B the SAME gives Q

- When both input A and B are the same, the XNOR logic gate produces a high output, produces a low output when both inputs A and B are different

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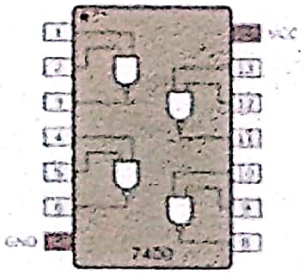
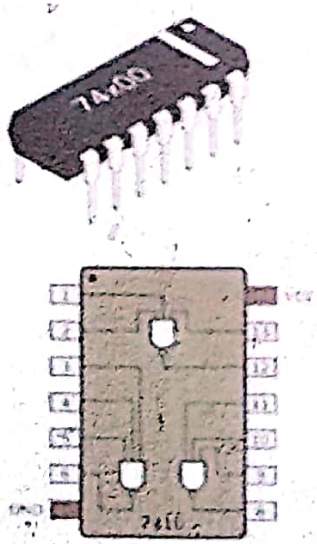
Logic Gates and Their ICs

- Logic Gate ICs are widely used in digital electronics, including computers, microprocessors, and other digital systems.

IC NUMBER	NAME OF THE GATE
7400, 7410	NAND
7402, 7427	NOR
7404, 7414	NOT Hex inverters
7408, 7411	AND
7432, 7433	OR
7486, 74136	EXOR/XOR
74266, 74280	EXNOR/XNOR



Quad 2-input NAND gate



Triple 3-input NAND gate

- The most common types of Logic Gate ICs are the TTL (Transistor-Transistor Logic) and CMOS (Complementary Metal-Oxide-Semiconductor) families.

For example: CD4011 is a CMOS equivalent for the 7400 (NAND gate)
CD4070 is a CMOS equivalent for the 7486 (XOR gate)

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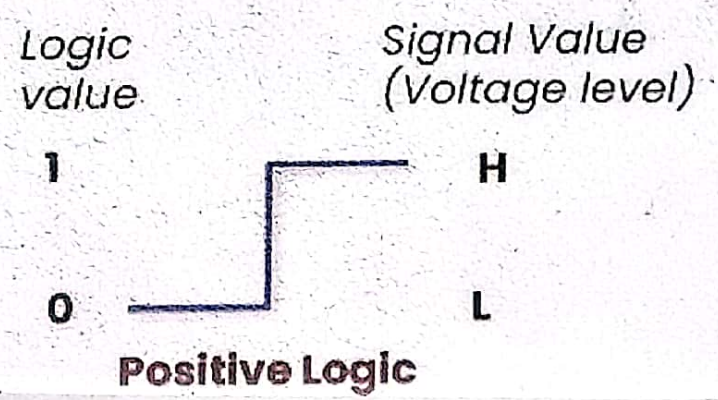
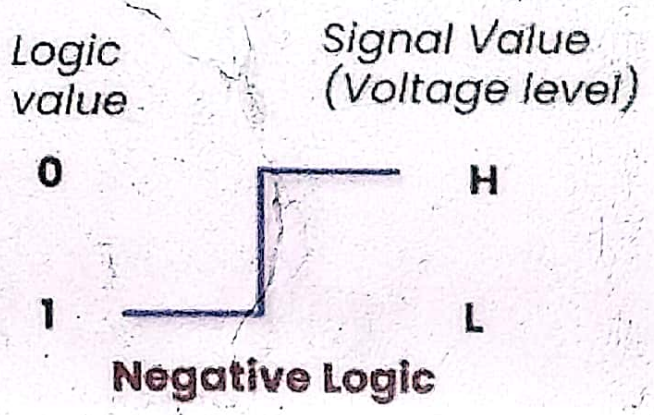
Digital Logic Families

❑ Positive and Negative Logic Systems

In **digital electronics**, binary logic is the foundation for representing information, where only two distinct states are used: '1' and '0'. These states are mapped to specific voltage levels, which can be interpreted as either positive logic or negative logic.

- In systems where a high signal level (H) corresponds to logical '1' and a low signal level (L) corresponds to logical '0', this configuration is referred to as **positive logic**. (eg- TTL, ECL family)
- In systems where a low signal level (L) corresponds to logical '1' and a high signal level (H) corresponds to logical '0', this configuration is referred to as **negative logic**. (eg- PMOS family)





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